

Python Programming and Machine Learning Lab

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

4038

Credits

4

Periods

5/week

Course Details

Title: Python Programming and Machine Learning Lab

Department: Common

Revision: REV_Python Programming and Machine Learning Lab

Pattern: Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 4038 — Python Programming and Machine Learning Lab**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

[illegible]

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

TABLE OF CONTENTS

Navigate handbook

Objectives & Outcomes

Module 1

Module 2

Module 3

Module 4

Model Question Paper

Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Python Programming and Machine Learning Lab.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3, CO2 → PO2: 3, CO3 → PO3: 3, CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Python Programming and Machine Learning Lab.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ഐക്യം ഐക്യം ഐക്യം ഐക്യം ഐക്യം
 ഐക്യം ഐക്യം ഐക്യം ഐക്യം ഐക്യം.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐക്യമേവ ജയിതം എന്നാണ് ഞങ്ങൾക്ക് മനസ്സിലായത്. ഐക്യമേവ ജയിതം.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ടെക്സ്റ്റ് ഫോർമാറ്റിംഗ് സാധനം ഉപയോഗിച്ച് ഈ സന്ദർഭം ഉപയോഗിക്കുക.
 Text: ടെക്സ്റ്റ് ഫോർമാറ്റിംഗ് സാധനം ഉപയോഗിച്ച് ഈ സന്ദർഭം ഉപയോഗിക്കുക.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ടെക്സ്റ്റ് ഫോർമാറ്റിംഗ് സാധനം ഉപയോഗിച്ച് ഈ സന്ദർഭം ഉപയോഗിക്കുക.
 Text: ടെക്സ്റ്റ് ഫോർമാറ്റിംഗ് സാധനം ഉപയോഗിച്ച് ഈ സന്ദർഭം ഉപയോഗിക്കുക.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Python Programming and Machine Learning Lab.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.